MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE JBN

This amendment forms a part of Military Specification MIL-M-28787/34B(EC), dated 14 January 1983, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, delete and substitute:

"1.2 Classification. This specification covers two classes of modules. Class I and class II as specified in MIL-STD-1389."

PAGE 2

3.3, delete and substitute:

"3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.19096 failures per million operating hours for class I or 0.47547 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3)."

3.4, delete and substitute:

"3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3)."

3.5.8, delete and substitute:

"3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/34-1 for class I and M28787/34-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)."

PAGES 4, 5, 6, 7, AND 8

TABLE I: Delete and substitute new table I as printed on pages 3 through 7.

PAGE 9

6.3, delete and substitute:

"6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

"6.3.1 Cross reference of classification. Military specification MIL-M-28787/34 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833446. Part number M28787/34-1 supersedes part numbers 2833446 and M28787/34."

FSC 5963

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PAGES 9 AND 10

6.4, delete and substitute:

"6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	285	285	mW
Supply voltage	5 ±10 percent	5 ±10 percent	V dc
Input voltage			
Low level	0 min, 0.8 max	0 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V
Input current, low level			
(V_{IL} = 0.0 V)	-1.73 max	-1.73 max	mA
(V_{IL} = 0.4 V)	-1.6 max	-1.6 max	mA
Input current, high level			
(V_{IH} = 2.4 V)	40 max	40 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA
Output voltage			
High level	2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
Low level	0 min, 0.4 max	0 min, 0.4 max	V
Stray capacitance	100 max	100 max	pF
Timing resistor (range)	5 min, 50 max	5 min, 50 max	kΩ
Output drive current, high level, maximum capability (V_{OH} = 2.4 V)	-0.72	-0.72	mA
Output drive current, low level, maximum capability (V_{OL} = 0.4 V)	9.6	9.6	mA
Propagation delay time			
t_{PLH}	5 min, 53 max	5 min, 53 max	ns
t_{PHL}	5 min, 54 max	5 min, 54 max	ns
Input setup time	40 min	40 min	ns
Input hold time	40 min	40 min	ns"

PAGE 12

6.5.3b.: Delete "0°C to 60°C" and substitute "-55°C to 85°C".

PAGE 24

TABLE VIII, test condition J1, unit column: Delete " " " and substitute "V".

PAGE 53

TABLE XV, test condition W, parameter column: Delete "Power supply capacitance" and substitute "Power supply filter capacitance".

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(Project 5963-N600)

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TABLE I. Module electrical requirements.

Test Condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	10	800	mV
B1-B8	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	0	1	mA
C	Isolation													
	Initial	V	S	10	∞	10	∞	10	∞	10	∞	10	∞	mΩ
	End-of-life	V	S	1	∞	1	∞	1	∞	1	∞	1	∞	mΩ
D1-D8	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E8	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F1-F8	Input current, low level	VII	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
G1-G4	Output short circuit current	VII	X	10	40	10	40	10	40	10	40	10	40	mA
H1-H4	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	0	400	mV
J1-J4	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K	Power supply current	IX	X	15	50	15	50	15	50	15	50	15	50	mA
L1, L2	Resistance													
	Initial	X	X	9,900	10,100	9,850	10,150	9,850	10,150	9,770	10,230	9,790	10,210	Ω
	End-of-life	X	X	9,800	10,200	9,785	10,215	9,785	10,215	9,705	10,295	9,725	10,275	Ω
L3, L4	Resistance													
	Initial	X	X	20,295	20,705	20,192	20,808	20,192	20,808	20,080	20,920	20,141	20,859	Ω
	End-of-life	X	X	20,090	20,910	20,059	20,941	20,059	20,941	19,875	21,125	19,936	21,064	Ω
L5, L6	Resistance													
	Initial	X	X	29,799	30,401	29,640	30,560	29,640	30,560	29,475	30,725	29,565	30,635	Ω
	End-of-life	X	X	29,498	30,702	29,452	30,748	29,452	30,748	29,174	31,026	29,264	30,936	Ω
L7, L8	Resistance													
	Initial	X	X	45,936	46,864	45,700	47,100	45,700	47,100	45,445	47,355	45,584	47,216	Ω
	End-of-life	X	X	45,472	47,328	44,742	48,028	44,742	48,028	44,487	48,288	44,626	48,144	Ω
L9, L10	Resistance													
	Initial	X	X	5,059	5,161	5,033	5,187	5,033	5,187	5,005	5,061	5,020	5,200	Ω
	End-of-life	X	X	5,008	5,212	5,000	5,220	5,000	5,220	4,977	5,094	4,987	5,233	Ω

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L11	Resistance													
	Initial	X	X	1,198	1,222	1,192	1,228	1,192	1,228	1,185	1,235	1,189	1,231	Ω
	End-of-life	X	X	1,186	1,234	1,174	1,246	1,174	1,246	1,167	1,253	1,171	1,249	Ω
L12	Resistance													
	Initial	X	X	2,465	2,515	2,453	2,527	2,453	2,527	2,439	2,541	2,450	2,530	Ω
	End-of-life	X	X	2,430	2,550	2,426	2,554	2,426	2,554	2,412	2,568	2,423	2,557	Ω
M1-M6	Capacitor dc leakage current	XI	X	0	3	0	3	0	3	0	3	0	3	µA
N1	Capacitance	XII	X	90	130	80	140	80	140	75	147	76	147	pF
N2	Capacitance	XII	X	1,080	1,320	960	1,440	960	1,440	912	1,512	912	1,512	pF
N3	Capacitance	XII	X	0.009	0.011	0.008	0.012	0.008	0.012	0.0076	0.0126	0.007	0.0126	µF
N4	Capacitance	XII	X	0.0738	0.0902	0.0656	0.0984	0.0656	0.0984	0.0623	0.1033	0.0623	0.1033	µF
N5	Capacitance	XII	X	0.612	0.748	0.544	0.816	0.544	0.816	0.526	0.857	0.526	0.857	µF
N6	Capacitance	XII	X	6.12	7.48	5.44	8.16	5.44	8.16	5.17	8.57	5.17	8.57	µF
N7	Capacitance	XII	X	1,080	1,320	960	1,440	960	1,440	912	1,512	912	1,512	pF
P1-P4	Capacitor dissipation factor	XII	X	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	%
P5,P6	Capacitor dissipation factor	XII	X	0	3	0	3	0	3	0	3	0	3	%
Q1-Q8	Pulse width	XIII	X	1.86	2.98	1.62	3.22	1.62	3.22	1.54	3.38	1.54	3.38	µs
Q9-Q16	Pulse width	XIII	X	1.72	2.74	1.50	2.96	1.50	2.96	1.42	3.11	1.42	3.11	µs
Q17,Q18, Q95,Q96	Pulse width	XIII	S	423	677	359	741	359	741	341	778	341	778	ns
Q19,Q20, Q97,Q98	Pulse width	XIII	S	847	1,353	721	1,479	721	1,479	685	1,553	685	1,553	ns
Q21,Q22, Q99,Q100	Pulse width	XIII	S	1,155	1,845	983	2,017	983	2,017	934	2,118	934	2,118	ns
Q23,Q24, Q101,Q102	Pulse width	XIII	S	1,733	2,787	1,470	3,050	1,470	3,050	1,397	3,203	1,397	3,203	ns
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See footnotes at end of table

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Test Condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q25,Q26 Q93,Q94	Pulse width	XIII	S	223	357	189	391	189	391	180	411	180	411	ns
Q27,Q28	Pulse width	XIII	S	3.43	5.47	2.92	5.98	2.92	5.98	2.77	6.28	2.77	6.28	µs
Q29,Q30	Pulse width	XIII	S	6.79	10.85	5.78	11.86	5.78	11.86	5.49	12.45	5.49	12.45	µs
Q31,Q32	Pulse width	XIII	S	9.9	15.7	8.4	17.2	8.4	17.2	7.98	18.1	7.98	18.1	µs
Q33,Q34	Pulse width	XIII	S	15.1	24.1	12.9	26.3	12.9	26.3	12.26	27.6	12.26	27.6	µs
Q35,Q36	Pulse width	XIII	S	14.5	23.1	12.4	25.2	12.4	25.2	11.8	26.5	11.8	26.5	µs
Q37,Q38	Pulse width	XIII	S	26.7	42.5	22.7	46.5	22.7	46.5	21.6	48.8	21.6	48.8	µs
Q39,Q40	Pulse width	XIII	S	52.8	84.2	44.9	92.1	44.9	92.1	42.7	96.7	42.7	96.7	µs
Q41,Q42	Pulse width	XIII	S	76.7	112.5	67.7	121.5	67.7	121.5	64.3	127.6	64.3	127.6	µs
Q43,Q44	Pulse width	XIII	S	117.2	187.2	99.7	204.7	99.7	204.7	94.7	214.9	94.7	214.9	µs
Q45,Q46	Pulse width	XIII	S	9.7	15.5	8.2	17.0	8.2	17.0	7.8	18.9	7.8	18.9	ms
Q47,Q48	Pulse width	XIII	S	17.9	28.7	15.2	31.4	15.2	31.4	14.5	33.0	14.5	33.0	ms
Q49,Q50	Pulse width	XIII	S	35.5	56.7	30.2	62.0	30.2	62.0	28.7	65.1	28.7	65.1	ms
Q51,Q52	Pulse width	XIII	S	51.6	82.4	43.9	90.1	43.9	90.1	41.7	94.6	41.7	94.6	ms
Q53,Q54	Pulse width	XIII	S	79	126	66.3	138.7	66.3	138.7	63.0	145.6	63.0	145.6	ms
Q55,Q56	Pulse width	XIII	S	46	101	40	111	40	111	38.0	116.6	38.0	116.6	ns
Q57,Q58	Pulse width	XIII	S	116	185	99	202	99	202	94.0	212.1	94.0	212.1	ns
Q59,Q60	Pulse width	XIII	S	216	344	184	376	184	376	174.8	394.8	174.8	394.8	ns
Q61,Q62	Pulse width	XIII	S	308	492	262	538	262	538	249	565	249	565	ns
Q63,Q64	Pulse width	XIII	S	477	763	405	835	405	835	385	877	385	877	ns
Q65,Q66 ---	Pulse width	XIII	S	3.17	5.07	2.70	5.54	2.70	5.54	2.56	5.82	2.56	5.82	µs

See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test Condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q67,Q68	Pulse width	XIII	S	6.27	10.01	5.33	10.95	5.33	10.95	5.06	11.50	5.06	11.50	µs
Q69,Q70	Pulse width	XIII	S	9.1	14.5	7.7	15.9	7.7	15.9	7.3	16.7	7.3	16.7	µs
Q71,Q72	Pulse width	XIII	S	13.9	22.3	11.8	24.4	11.8	24.4	11.2	25.6	11.2	25.6	µs
Q73,Q74	Pulse width	XIII	S	117	187	100	204	100	204	95	214	95	214	µs
Q75,Q76	Pulse width	XIII	S	216	346	184	378	184	378	175	397	175	397	µs
Q77,Q78	Pulse width	XIII	S	428	684	364	748	364	748	346	785	346	785	µs
Q79,Q80	Pulse width	XIII	S	622	994	528	1,088	528	1,088	502	1,142	502	1,142	µs
Q81,Q82	Pulse width	XIII	S	952	1,520	810	1,662	810	1,662	770	1,745	770	1,745	µs
Q83,Q84	Pulse width	XIII	S	0.97	1.55	0.82	1.70	0.82	1.70	0.78	1.79	0.78	1.79	ms
Q85,Q86	Pulse width	XIII	S	1.79	2.87	1.52	3.14	1.52	3.14	1.44	3.30	1.44	3.30	ms
Q87,Q88	Pulse width	XIII	S	3.55	5.67	3.02	6.08	3.02	6.08	2.87	6.38	2.87	6.38	ms
Q89,Q90	Pulse width	XIII	S	5.16	8.24	4.39	9.01	4.39	9.01	4.17	9.46	4.17	9.46	ms
Q91,Q92	Pulse width	XIII	S	7.89	12.61	6.71	13.79	6.71	13.79	6.37	14.48	6.37	14.48	ms
R1-R4	Transition time (t_{TLH})	XIII	S	10	80	10	85	10	85	10	85	10	85	ns
S1-S4	Transition time (t_{THL})	XIII	S	2	25	2	30	2	30	2	30	2	30	ns
T1-T16	Propagation delay time (t_{PLH})	XIII	S	5	49	5	53	5	53	5	53	5	53	ns
U1-U16	Propagation delay time (t_{PHL})	XIII	S	5	49	5	54	5	54	5	54	5	54	ns
V1-V8	Retrigger pulse width	XIV	S	228	406	199	435	199	435	189	457	189	457	ns
V9-V16	Retrigger pulse width	XIV	S	1.72	2.74	1.50	2.96	1.50	2.96	1.43	3.11	1.43	3.11	µs
W	Power supply filter capacitance	XV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
X1-X4 ---	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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Test Condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y4	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
AA1-AA8	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	2	15	pF
BB1,BB2	Stray capacitance	XVIII	Q	0	100	0	100	0	100	0	100	0	100	pF
BB3,BB4	Stray capacitance	XVIII	Q	0	50	0	50	0	50	0	50	0	50	pF
BB5	Stray capacitance	XVIII	Q	5	30	5	30	5	30	5	30	5	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be saple acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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